Article

Stress-Testing Data Minimization Across Review Budgets and Decision Stages in Educational Early Warning: Paired Evidence from OULAD and Two UCI Datasets

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Abstract

Data-minimization claims for educational early-warning systems can change with review capacity, prediction stage, excluded field block, and evaluation metric. We extended a provenance-corrected analysis of the Open University Learning Analytics Dataset (OULAD) and two separately identified UCI datasets, re-evaluating frozen legacy models at 5%, 10%, 20%, and 30% within-group review budgets without refitting them. We also tested whether directly excluded sex/gender and disability/special-needs fields remained predictable from retained inputs, and fitted only the histogram-gradient-boosting models needed for two operationally harmonized blocks – sex/gender and socioeconomic/family information – accepting new restricted models only when a same-run full control reproduced its frozen probability to within 10^{-12}. Paired bootstraps used 5,000 primary replicates with Bonferroni-percentile intervals across all frozen stages and budgets. Of 296 primary precision cells, 30 had family-wise intervals below zero and none above zero; this count includes planned duplicate views and is not 30 independent confirmations. Sex/gender exclusion showed no family-wise workload-precision difference in any dataset, whereas socioeconomic/family exclusion reduced global predictive performance at every OULAD horizon and both UCI 697 stages, with workload penalties concentrated at selected OULAD 30% queues and large at UCI 697 enrollment; UCI 320 workload differences remained inconclusive. Fixed probes predicted excluded fields above chance, establishing residual predictability rather than actual proxy use by the outcome models. The results support stage-, capacity-, block-, and metric-specific governance rather than a universal claim that direct exclusion is cost-free, and are post-development, conditional on fitted models, without establishing equivalence, fairness, privacy, or causal educational benefit.

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1. Introduction

Educational early-warning systems rank students for possible review or support. Their practical value is determined not only by discrimination but also by the operating queue: how many registrations staff can review and how many selected registrations later experience the target outcome. Precision–recall measures are therefore relevant when risk prevalence and review capacity are constrained (Saito & Rehmsmeier, 2015). A retrospective ranking study nevertheless remains distinct from an intervention study. Field experiments are needed to determine whether displaying alerts changes counselling, teaching practice, retention, or achievement (Plak et al., 2022).

Background variables can improve prediction while creating questions about necessity, proportionality, and governance. Data minimization is a lifecycle design obligation, not a claim that deleting a direct field automatically makes a model fair, private, or lawful (Drachsler & Greller, 2016; European Parliament and Council of the European Union, 2016; Rets et al., 2023). Direct exclusion can leave proxies and can change which individuals are selected even when aggregate precision is similar. Its cost should therefore be evaluated at the intended decision unit and workload.

Capacity-based alerting, calibration, subgroup audits, repeated landmarks, and alert-list stability are established evaluation components rather than novel contributions (Wu et al., 2026; Wu & Luo, 2026). Leakage-excluded OULAD protocols emphasize cutoff-first truncation and feature-availability audits (Le et al., 2026); the present analysis shares that temporal concern but asks a different question about workload-constrained field exclusion. Privacy-conscious engagement-only prediction and causal or adversarial analyses of demographic data further show that predictive utility, privacy, and fairness cannot be collapsed into one claim (Chiu et al., 2026; Hasan & Fritz, 2022). Our contribution is a post-development stress test of exclusion claims across four review budgets, a held-out audit of residual predictability after direct field deletion, two operationally harmonized block comparisons with exact same-run controls, and paired family-wise uncertainty across all frozen stages and budgets. OULAD, UCI 697, and UCI 320 remain separately identified and are not pooled as replications.

The OULAD data, temporal split, model families, workload framing, demographics, and interpretability have been used in an earlier article by overlapping investigators (Shaikhanova et al., 2025). They are not claimed as new. The original question here concerns exclusion of all six background fields at day 42. The gender/disability-only policy and the additional horizons were defined after earlier OULAD outcomes had been inspected. They are secondary, post-development analyses. The study asks:

1. Do restricted-minus-full workload differences retain their sign and practical scale at 5%, 10%, 20%, and 30% review budgets?
2. Are directly excluded sex/gender and disability/special-needs fields still predictable from the retained feature view in held-out data?
3. How do the operationally harmonized sex/gender and socioeconomic/family blocks behave across datasets and stages when every new restricted model has a same-run full control?
4. Which conclusions survive paired family-wise uncertainty across all frozen stages and budgets?

5. Do the earlier complete-exclusion, targeted-exclusion, and click-independent findings remain appropriately bounded after this stress test?

2. Materials and Methods

2.1. Analysis status and provenance correction

The analysis is a locked-after-audit, post-development exploratory reanalysis. The 2014J outcomes had been inspected in earlier development. The -0.03 complete-exclusion reference was selected after earlier results, and the gender/disability-only policy was not part of the original v0.2.1 question. During forensic review, main models for days 14–56 on the official archive were also observed before the complete C12 production protocol was fixed. After that diagnostic step, no policy, horizon, feature recipe, model grid, seed, alert rule, bootstrap plan, or gate was changed. This chronology prevents a confirmatory or independent-validation interpretation.

The official OULAD archive had SHA-256 `f2ed1902616c1fe8d2824d872c0b7d2d7 2be435bf0124d077044fe4be2c6d3e4`. All archive members passed CRC validation and 54 input checks. A superseded recovery archive had SHA-256 `b344faabbf728c2fec939 13127813391d259b959aa283de715db51b2c0eaa81b` and replaced 1,111 literal `?` values in `imd_band` with empty fields. Under the inherited preprocessing recipe, the literal marker was an explicit categorical level whereas empty fields were most-frequent-imputed. An exact replay of v0.2.1 on the official archive reproduced all 64 saved workload rows. This established a concrete provenance error rather than evidence that the intermediate numerical results were fabricated. The superseded run is retained only as audit history.

2.2. Dataset, outcome, and temporal partitions

OULAD contains 32,593 student–module–presentation registrations across 22 presentations of seven anonymized modules, assessment records, and 10,655,280 virtual-learning-environment records (Kuzilek et al., 2017). At landmark day t, the dynamic risk set contained registrations that had begun by t and had no withdrawal recorded on or before t. The binary outcome was future Fail or Withdrawn versus Pass or Distinction. Models used 2013B and 2013J for fitting, 2014B for model selection and calibration, and 2014J for temporal evaluation. Snapshots were created after the end of days 14, 28, 42, 56, 70, and 90.

Because withdrawals remove registrations from later risk sets, cohort size, composition, and prevalence change over time. Cross-landmark differences therefore do not estimate an optimal intervention time. A common-registration sensitivity retained registrations present at all first four horizons or all six horizons, but this conditions on later retention and is also non-causal.

2.3. As-of-time features and field-exclusion policies

Virtual-learning-environment events were aggregated only when their event day was no later than the landmark. Features described cumulative and recent click volume, active days, timing, inactivity, pre-start activity, and click trend. An assessment submission counted only when it was non-banked and both its due date and submission date were no later than the landmark. Scores were excluded because OULAD does not provide grading or score-release timestamps. Expected-assessment denominators were limited to assessments due by the landmark.

All standard policies used module, previous attempts, studied credits, registration lead time, and the available activity and assessment-process features. The *full* policy additionally used gender, region, highest education, Index of Multiple Deprivation band, age band, and disability. The *six-field-excluded* policy removed all six. The secondary *gender/disability-*

excluded policy removed only gender and disability. Six leave-one-field-out models at days 14 and 42 removed one field at a time. Excluded fields remained available for descriptive auditing; neither candidate is a claim of complete lifecycle minimization.

2.4. Models, selection, and calibration

Histogram gradient boosting was implemented in scikit-learn (Pedregosa et al., 2011). Numeric variables were median-imputed with missingness indicators. Categorical variables were most-frequent-imputed and ordinal-encoded, with transformations fitted on training data only. The number of leaves was selected from $\{7, 15, 31\}$ by 2014B average precision, breaking validation ties in favor of the simpler model. Other settings were learning rate 0.05, at most 250 iterations, minimum leaf size 30, $L_2 = 1$, disabled internal early stopping, and seed 42. A positive-slope Platt model was fitted on 2014B predictions and transferred unchanged to 2014J. Calibration was evaluated rather than assumed (Collins et al., 2024; Moons et al., 2025; Van Calster et al., 2019).

The production run fitted 105 validation candidates and saved 36 selected pipelines before uncertainty analysis. It also saved 36 validation and 36 test prediction files. All new contrasts compare policies within this same official-data run.

2.5. Workload-constrained estimand and ties

Within each module m, calibrated risks were sorted and

$$k_m = \lceil 0.10 n_m \rceil$$

registrations were selected. Micro-aggregated precision and recall were

$$\text{Precision@10\%} = \frac{\sum_m TP_m}{\sum_m k_m}, \qquad \text{Recall@10\%} = \frac{\sum_m TP_m}{\sum_m N_{+,m}}.$$

Exact score ties were resolved by a SHA-256 key derived from module, presentation, and student identifier. A sensitivity calculated expected yield under uniform ordering within exact boundary-score blocks. This explicit rule accounts for a two-alert discrepancy in a prior day-28 replay and is part of the estimand, not a model improvement.

Average precision, AUROC, Brier score, log loss, and ten-bin expected calibration error supplemented the capacity metric. The unchanged original gate required the pointwise lower 95% bound for six-field-excluded minus full day-42 Precision@10% to exceed -0.03. This is a post-development operational reference, not a formal non-inferiority margin.

2.6. Paired uncertainty and common-cohort sensitivity

Student identifiers were resampled with replacement, retaining all selected registrations and duplicate multiplicities. Each replicate recomputed both module-specific queues. Day 42 used 5,000 paired cluster-bootstrap replicates; each other dynamic horizon used 2,000. Models, transformations, hyperparameters, and calibrators were not refitted inside the bootstrap. Percentile and basic intervals were recorded. Bonferroni-percentile sensitivity intervals were calculated over each six-horizon policy family. A two-sided sign proportion was retained only as a descriptive bootstrap statistic, not a confirmatory p-value.

The four-horizon common cohort contained 8,853 registrations from 8,536 students; the six-horizon cohort contained 8,507 registrations from 8,216 students. Within each, day-42-minus-day-14 difference-in-differences compared the change in each exclusion gap with the full-control change. These analyses condition on later registration retention.

2.7. Click-independent tiers and unchanged strict gate

At day 42, the original strict tier excluded all click/activity fields and retained context plus due assessment-process features. The enriched tier added typed, due-by-landmark process counts and ratios without using scores, click telemetry, or future records. A typed alternative changed the process representation. Each tier selected complexity on 2014B and was calibrated on 2014B. An expanded leaf grid added 63 and 127 leaves; it produced the same selected predictions as the standard grid.

The unchanged strict gate required click-independent clean Precision@10% to be no more than 0.05 below the same-run six-field-excluded standard comparator. Improvement over the original strict tier could not substitute for failure against this comparator.

2.8. External UCI datasets and cross-fitted controls

Two official UCI archives were added after the OULAD analysis and are reported as distinct external analyses. UCI 697, *Predict Students' Dropout and Academic Success*, contains 4,424 higher-education students, 36 predictors, and the outcomes Dropout, Enrolled, and Graduate (Realinho et al., 2021). The primary terminal-status comparison used Dropout versus Graduate and excluded Enrolled; a secondary outcome treated Graduate and Enrolled as non-dropout. Enrollment models used 24 enrollment-time fields. Semester-one models added six first-semester curricular-unit fields; all second-semester fields were prohibited. The historical *legacy operational* subset retained application order, course, attendance schedule, previous qualification and grade, and admission grade, adding first-semester fields at the later stage. Further same-run policies excluded direct personal fields, family/financial fields, or gender plus educational special-needs status.

UCI 320, *Student Performance*, provides separate Mathematics ($n = 395$) and Portuguese ($n = 649$) samples from two secondary schools (Cortez, 2008). The subjects were not pooled: the supplied 13-field linkage key was not unique. The target was final-grade failure ($G3 < 10$). Baseline models excluded G1, G2, and G3; period-one models added G1; period-two models added G1 and G2 while always excluding G3. Same-run policies removed sex, a demographic/family block, a sensitive/wellbeing block, or retained only school-observable fields. These labels describe frozen operational sets rather than legal or ethical classifications of individual fields.

Both analyses used ten shared stratified outer folds. Within each outer training fold, a stratified 20% split was used for HGB leaf selection from $\{7, 15, 31\}$ and Platt calibration. Logistic regression was a sensitivity model. Every selected fold model, calibrator, split assignment, and out-of-fold probability was saved before statistics. Alert budgets were allocated separately within Course for UCI 697 and school for UCI 320, using an explicit SHA-256 row tie-break. Paired row bootstraps used 5,000 replicates for primary HGB contrasts and 2,000 for secondary contrasts. Bonferroni intervals covered the two UCI 697 or three UCI 320 stages within each frozen policy family. Because earlier UCI 697 results existed and all external choices followed OULAD inspection, neither dataset is presented as independent confirmation.

2.9. C14 multi-budget extension and frozen status

The C14 extension was specified after C12 and C13 results were known and is not preregistered confirmation. Before C14 results were computed, the protocol fixed four review fractions, $b \in \{0.05, 0.10, 0.20, 0.30\}$. Within each operational group, the queue contained $\lceil bn_g \rceil$ observations. OULAD retained its module-specific SHA-256 tie rule; UCI 697 and UCI 320 retained the C13 Course- and school-specific rules. C14B recomputed queue size, true and false alerts, precision, recall, lift, alert-set Jaccard similarity, and the fraction of full-model alerts retained, using saved C12 test probabilities and C13 out-of-fold probabilities only. No model, feature recipe, calibrator, or split was altered.

2.10. Proxy-persistence audit

C14C asked whether an excluded field remained predictable from the fields retained by its targeted-exclusion view. It did not ask whether an outcome model actually used a particular proxy. Fixed logistic probes ($C = 1$, 2,000 maximum iterations) were primary; fixed HGB probes with 15 leaves were nonlinear sensitivities. OULAD probes targeted gender and disability, were fitted on 2013B/2013J, used 2014B only to select the balanced-accuracy threshold, and were evaluated on 2014J; an unseen-student subset excluded 2014J students observed during fitting. UCI 697 probes targeted Gender and Educational special-needs status, and UCI 320 probes targeted sex, using the ten saved outer folds and preprocessing within each fit fold. Outcomes, identifiers, split fields, the target field itself, and future-stage fields were prohibited. We report AUROC, average precision, its lift over target prevalence, balanced accuracy, Brier score, prevalence, and positive-case counts. These probes are descriptive predictability tests, not fairness tests, privacy attacks, or causal analyses.

2.11. Operationally harmonized exclusion blocks

C14D fixed two cross-dataset block labels. The *sex/gender* block comprised gender in OULAD, Gender in UCI 697, and sex in UCI 320. The *socioeconomic/family* block comprised highest education and area-deprivation band in OULAD; parental qualification and occupation, debtor and tuition status, and scholarship status in UCI 697; and parental education and occupation in UCI 320. These labels describe operational groupings, not measurement equivalence or a shared latent construct.

Exact saved same-run pairs were reused wherever available. New HGB outcome models were fitted only for OULAD no-gender at days 28, 56, 70, and 90; OULAD socioeconomic/family exclusion at all six horizons; and UCI 320 socioeconomic/family exclusion for both subjects at all three stages. Each of the 12 distinct stage-specific new full controls was refitted in the same C14D execution. A pair was admissible only if its control reproduced the corresponding frozen full probability with maximum absolute error no greater than 10^{-12}. No new logistic or UCI 697 outcome model was fitted. The canonical XuetangX C05 archive was neither extracted nor retrained and remained excluded from demographic claims.

2.12. C14 paired uncertainty and multiplicity

C14E recomputed queues within each bootstrap replicate. OULAD resampled student identifiers and retained all registration multiplicities; UCI resampled rows. The same multiplicity vector was used for full and restricted predictions and all four budgets. A deterministic stream at dataset, subject/outcome, and stage level was also shared across exact C14B/C14D duplicate views. Primary HGB contrasts used 5,000 replicates and sensitivity contrasts used 2,000. For workload metrics, each family was defined by dataset, subject/outcome, named policy or block, model, and grid and covered every frozen stage times four budgets; probability metrics covered stages only. We report pointwise 95% percentile intervals and Bonferroni-percentile family-wise intervals. Bootstrap sign proportions and their Bonferroni products are descriptive paired statistics rather than randomization-test p-values. The primary estimand was precision restricted minus full; recall, overlap, AP, AUROC, and Brier differences were secondary. Failure to exclude zero was not treated as equivalence.

2.13. Computational verification

The official OULAD input passed 54 of 54 checks. OULAD model and prediction artifacts passed 861 of 861 checks; bootstrap replay, common-cohort identities, strict-grid equivalence, gates, and the evidence ledger passed 1,954 of 1,954 checks. For C13, 114 UCI model bundles and 243,220 out-of-fold rows passed complete hash, split-disjointness, leakage, completeness, and prediction-replay checks; the maximum replay error was 1.11×10^{-16}. C14B reproduced 150 saved configurations at the 10% operating point with maximum error 2.22×10^{-16}. C14C independently replayed 44 probes and 250,044 predictions. C14D independently replayed 28 new bundles; its same-run full-control maximum error was 1.11×10^{-16}, and 30 harmonized pairs contained 274,004 paired prediction rows. C14E independently recalculated all 2,869 interval rows from 42 hashed draw files. These are internal reproducibility checks, not external validation.

3. Results

3.1. The official-data rerun changed some intermediate results but restored provenance

The official-data run reproduced the v0.2.1 six-field-excluded precision values at days 14, 42, and 56 and, under the explicit SHA tie-break, selected 762 rather than 760 true-risk alerts at day 28. Full-model true-alert counts differed from the superseded recovery run by $+3$, -9, -1, $+1$, -1, and -4 across days 14–90. These changes were not adjusted toward historical values. The exact transition is reported in the supplement.

At day 42, the active 2014J risk set contained 9,040 registrations and 3,584 future Fail/Withdrawn outcomes. Each standard policy generated 908 alerts. Full selected 769 true-risk registrations, six-field exclusion selected 762, and gender/disability exclusion selected 765.

3.2. Original complete-exclusion result: day-42 gate passed, six-horizon stability did not

At the original day-42 operating point, six-field-excluded Precision@10% was 0.8392 versus 0.8469 for the full model. The paired difference was -0.0077 (95% percentile interval $[-0.0265, 0.0155]$). Its lower bound exceeded the unchanged -0.03 reference, so the original exploratory gate passed.

This result did not generalize uniformly across horizons. At day 14, six-field exclusion selected 670 true-risk registrations among 935 alerts versus 708 for full: a precision difference of -0.0406 with a pointwise interval $[-0.0608, -0.0107]$ and a six-horizon Bonferroni interval $[-0.0692, -0.0036]$. Average precision was lower for six-field exclusion at every horizon. Complete exclusion therefore cannot be described as generally lossless, even though its original day-42 gate passed.

3.3. Secondary gender/disability exclusion preserved queue precision within the exploratory reference

At day 42, gender/disability-excluded precision was 0.8425 versus 0.8469 for full, a difference of -0.0044 (Table 1). The pointwise interval included zero. AUROC and Brier score nevertheless favored full by small amounts, showing that near-equal queue precision did not imply equal global ranking or calibration performance.

Table 1. Official-data C12 day-42 comparison of gender/disability exclusion with the same-run full control.

Metric	G/D excluded	Full	Difference	95% interval
Alerts	908	908	—	—
True-risk alerts	765	769	−4	—
False alerts	143	139	+4	—
Precision@10%	0.8425	0.8469	-0.0044	[-0.0110, 0.0099]
Recall@10%	0.2134	0.2146	-0.0011	[-0.0028, 0.0025]
Average precision	0.6934	0.6943	-0.0009	[-0.0022, 0.0005]
AUROC	0.7585	0.7603	-0.0018	[-0.0027, -0.0009]
Brier score	0.1915	0.1909	+0.0006	[0.0003, 0.0008]

Intervals are paired student-cluster percentile intervals with 5,000 replicates, conditional on locked models. Positive Brier differences favor the full model.

Across all six horizons, precision differences ranged from −0.0044 at day 42 to +0.0035 at day 90 (Table 2; Figure 1). Every pointwise interval included zero. All six Bonferroni lower bounds exceeded −0.03; the minimum was −0.0173 at day 28. This is a post-development sensitivity compatible with a precision loss smaller than the reference under the stated conditional bootstrap. It is not a formal non-inferiority result.

Table 2. Gender/disability-excluded versus full Precision@10% at six dynamic landmarks.

Day	Alerts	Candidate TP	Candidate P	Full TP	Full P	Difference	Pointwise 95%	Bonferroni 95%
14	935	709	0.7583	708	0.7572	+0.0011	[-0.0107, 0.0128]	[-0.0149, 0.0171]
28	923	763	0.8267	764	0.8277	-0.0011	[-0.0108, 0.0141]	[-0.0173, 0.0195]
42	908	765	0.8425	769	0.8469	-0.0044	[-0.0110, 0.0099]	[-0.0143, 0.0134]
56	890	803	0.9022	801	0.9000	+0.0022	[-0.0067, 0.0124]	[-0.0101, 0.0165]
70	873	813	0.9313	815	0.9336	-0.0023	[-0.0092, 0.0069]	[-0.0115, 0.0092]
90	855	804	0.9404	801	0.9368	+0.0035	[-0.0082, 0.0070]	[-0.0117, 0.0093]

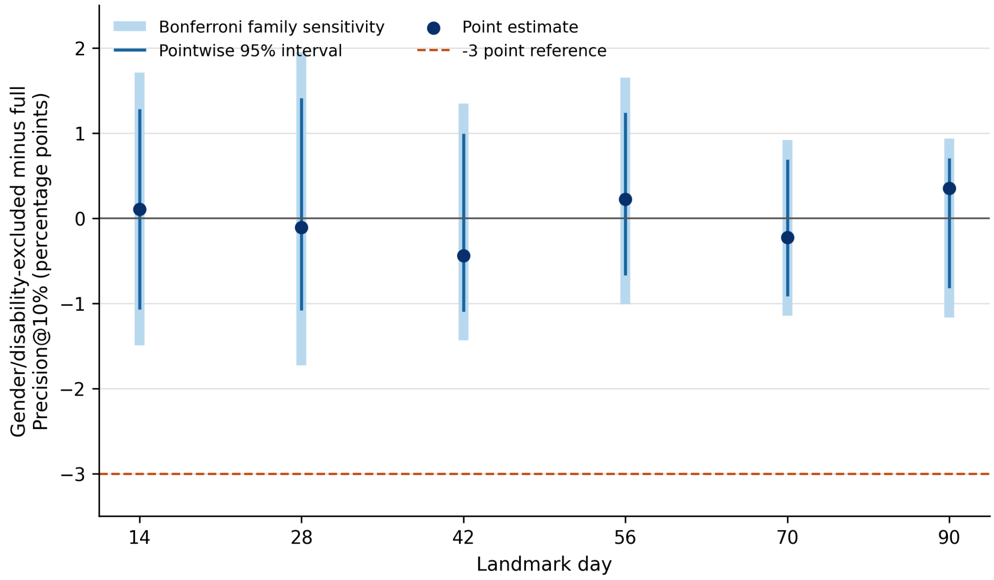

Figure 1. Paired gender/disability-excluded minus full Precision@10% contrasts. Dark intervals are pointwise student-cluster percentile intervals; pale intervals are Bonferroni family sensitivities. The dashed line is the post-development −3-percentage-point reference. Models were locked before resampling.

The favorable precision sensitivity was metric-specific. At day 14, average precision decreased by −0.00342 (Bonferroni interval [−0.00613, −0.00071]). At the later five horizons, the family-wise average-precision intervals included zero. At day 42, the small AUROC difference was −0.0018 (pointwise interval [−0.0027, −0.0009]) and the Brier difference was +0.0006 ([0.0003, 0.0008]), both favoring full.

Near-equal aggregate precision also did not imply identical recipients. Relative to full, gender/disability exclusion replaced 54, 61, 37, 35, 21, and 17 alerts in each direction over days 14–90; Jaccard similarity increased from 0.8908 to 0.9610. Complete exclusion replaced 290 alerts per direction at day 14 and 169 at day 42. Recipient-level review is therefore necessary even when a queue-level difference is small.

3.4. Leave-one-field-out and common-cohort analyses did not establish a causal field or timing rule

The largest leave-one-field-out precision loss occurred when highest education was removed at day 14: −0.0342, pointwise interval [−0.0544, −0.0085], with a Bonferroni lower bound of −0.0619. At day 42, all six point differences lay between −0.0066 and +0.0077. These contrasts can identify operational sensitivity but cannot rank fields by causal importance.

For gender/disability exclusion, the day-42-minus-day-14 difference-in-differences was 0.0000 in the four-horizon common cohort (95% interval [−0.0169, 0.0180]) and +0.0070 in the six-horizon cohort ([−0.0129, 0.0234]). Complete exclusion showed positive pointwise changes, but the Bonferroni sensitivity intervals crossed zero. Conditioning on later retention and the changing information set prevent a monotonic or causal timing interpretation.

3.5. Enriched click-independent features improved the original strict tier but failed the unchanged gate

The enriched due-process tier increased average precision by 0.0256 relative to the original strict tier (95% interval [0.0191, 0.0322]), increased AUROC by 0.0315 ([0.0234, 0.0397]), and reduced Brier score by 0.0070. Its Precision@10% increase was only +0.0055 ([−0.0188, 0.0210]). Point metrics are shown in Table 3.

Table 3. Official-data C12 day-42 click-independent tier results.

Policy	Leaves	True alerts	Precision@10%	AP
Six-field-excluded standard	7	762	0.8392	0.6798
Strict original	31	669	0.7368	0.5883
Strict enriched due-process	7	674	0.7423	0.6139
Strict typed alternative	7	670	0.7379	0.5945

Against the same-run six-field-excluded standard comparator, enriched strict precision was 0.7423 versus 0.8392: difference -0.0969 (95% interval $[-0.1289, -0.0738]$). The unchanged -0.05 gate therefore failed. The typed alternative also failed, and the expanded grid produced identical predictions. The inherited count of 712 enriched true alerts was not recovered; C12 selected 674. The inherited C06/C08 value remains `NOT_REVERIFIED` and is not used as evidence.

3.6. UCI 697 reproduced a large enrollment-time penalty that attenuated after semester one

For the primary Dropout-versus-Graduate outcome, the full HGB control attained Precision@10% of 0.9111 at enrollment and 0.9865 after semester one. The historical legacy-operational subset attained 0.5849 and 0.9730, respectively (Table 4). The restricted-minus-full difference therefore changed from -32.61 percentage points (two-stage Bonferroni interval $[-39.25, -27.06]$) to -1.35 points ($[-4.04, 0.00]$). The paired attenuation estimate was $+31.27$ points (95% interval $[25.75, 36.76]$). The old workload point estimates were reproduced exactly; under the new canonical paired bootstrap, however, the later interval touched zero rather than remaining strictly negative.

Table 4. UCI 697 HGB workload precision for the primary terminal-status outcome.

Stage	Full	Legacy operational	Difference (points)	Family-wise 95% CI
Enrollment	0.9111	0.5849	-32.61	[-39.25, -27.06]
After semester 1	0.9865	0.9730	-1.35	[-4.04, 0.00]

The field blocks were not interchangeable. At enrollment, removing family/financial fields changed Precision@10% by -21.83 points, whereas removing seven direct personal fields changed it by -0.54 points and removing gender plus special-needs status changed it by $+0.81$ points; the latter two workload intervals included zero. Yet direct-personal exclusion reduced AP by 0.0091 and AUROC by 0.0108 with family-wise intervals below zero. Targeted gender/special-needs exclusion reduced enrollment AUROC by 0.0059, also with a family-wise interval below zero. After semester one, workload differences were small, but AP and Brier still identified selected losses. Thus, attenuation at the queue boundary did not make all restricted models equivalent.

3.7. UCI 320 confirmed grade-stage information but not a stable exclusion penalty

The UCI 320 full controls improved sharply when grades became available (Table 5; Figure 2). In Mathematics, AP increased from 0.432 at baseline to 0.825 with G1 and 0.941 with G1+G2; Precision@10% increased from 0.575 to 0.925 and 0.975. Portuguese AP increased from 0.412 to 0.663 and 0.845, while Precision@10% increased from 0.409 to 0.591 and 0.727.

Table 5. UCI 320 HGB full-control performance by grade availability; subjects are not pooled.

Subject	Stage	AP	AUROC	Precision@10%
Mathematics	Baseline	0.432	0.618	0.575
Mathematics	+G1	0.825	0.906	0.925
Mathematics	+G1+G2	0.941	0.971	0.975
Portuguese	Baseline	0.412	0.796	0.409
Portuguese	+G1	0.663	0.930	0.591
Portuguese	+G1+G2	0.845	0.959	0.727

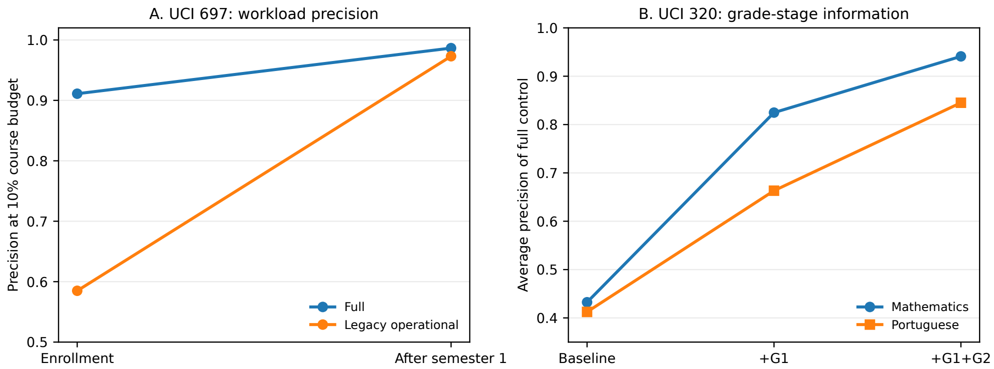

Figure 2. Distinct external stage profiles. Panel A shows UCI 697 full and legacy-operational HGB workload precision. Panel B shows UCI 320 full-control HGB average precision as G1 and G2 become available; Mathematics and Portuguese are analyzed separately. The panels are not pooled estimates or direct replications.

The restricted-minus-full findings were much less stable. Nearly all UCI 320 HGB pointwise and within-policy family-wise intervals included zero for AP, AUROC, Brier, and workload precision. The sole HGB contrast whose three-stage family-wise interval excluded zero was a small Mathematics period-one AUROC improvement after removing sex (+0.0050). Because this isolated result sits among multiple subjects, policies, and metrics, it is retained as a non-confirmatory finding. UCI 320 therefore supports the informational value of staged grades, not a precise or universal exclusion-cost trajectory.

3.8. Budget sensitivity changed which workload conclusions survived

Across C14B, 44 primary stage–policy contrasts yielded 176 precision cells at four budgets. The OULAD minimized policy was negative in 19 of 24 cells but crossed zero in both directions across horizons and budgets. Its family-wise precision interval was below zero only at the 30% budget on days 14, 28, and 56 (Table 6). The targeted gender/disability policy had no family-wise workload difference in any of its 24 cells. Thus the original 10% operating point did not summarize the larger queues: a policy that was inconclusive at one boundary could incur a detectable loss when review extended deeper into the ranked list.

In UCI 697, the legacy operational policy remained substantially below full at enrollment for all four budgets, from −34.92 percentage points at 5% to −17.97 points at 30%. Semester-one family-wise intervals were below zero only at 20% and 30%. The no-family/financial policy showed the same family-wise pattern. No primary UCI 320 workload cell excluded zero after correction. These results preserve the earlier stage interpretation but qualify it: attenuation after stronger academic evidence was incomplete and depended on the queue boundary.

Table 6. Selected primary restricted-minus-full precision differences across review budgets. Values are percentage points. Asterisks mark Bonferroni-percentile family-wise intervals that exclude zero.

Context	5%	10%	20%	30%
OULAD day14, minimized	-3.62	-4.06	-2.19	-2.07*
OULAD day28, minimized	+0.22	-0.22	-1.52	-1.99*
OULAD day42, minimized	+1.75	-0.77	-0.50	-1.51
OULAD day56, minimized	-0.22	+0.34	-0.96	-1.99*
OULAD day70, minimized	+0.68	-0.46	-0.69	-1.53
OULAD day90, minimized	-0.23	+0.94	-0.82	-0.43
UCI 697 enrollment, legacy	-34.92*	-32.61*	-22.37*	-17.97*
UCI 697 semester1, legacy	-3.17	-1.35	-3.55*	-3.56*

3.9. Excluded fields remained predictable from retained inputs

Every fixed primary logistic probe achieved held-out AUROC above 0.63 (Table 7; Figure 3). OULAD gender AUROC ranged from 0.690 to 0.812 across horizons and disability from 0.663 to 0.680. Sex AUROC ranged from 0.746 to 0.750 in UCI 320 Mathematics and from 0.734 to 0.734 in Portuguese. UCI 697 Gender reached 0.771–0.774. Educational special-needs status in UCI 697 was less stable: AUROC was 0.637–0.655, but only 40 positive cases occurred in each outer-fold evaluation aggregation and average precision was approximately 0.020 at prevalence 0.011. The nonlinear HGB sensitivity was weaker for this rare target.

The OULAD unseen-student sensitivity did not remove the signal: gender AUROC ranged from 0.712 to 0.818 and disability from 0.659 to 0.676. These results show that retained variables contain information associated with the deleted fields. They do not identify the source of that association, demonstrate that the risk model used it, or establish a fairness or privacy violation.

Table 7. Primary fixed logistic probes for directly excluded fields. Ranges cover frozen stages; AP lift is average precision divided by target prevalence.

Dataset	Target	Stages	Positive cases	AUROC range	AP-lift range
OULAD	disability	6	761–868	0.663–0.680	1.52–1.68
OULAD	gender	6	4905–5425	0.690–0.812	1.31–1.43
UCI 320 Math.	sex	3	187	0.746–0.750	1.50–1.50
UCI 320 Port.	sex	3	266	0.734–0.734	1.63–1.64
UCI 697	Educational special needs	2	40	0.637–0.655	1.77–1.88
UCI 697	Gender	2	1249	0.771–0.774	1.89–1.90

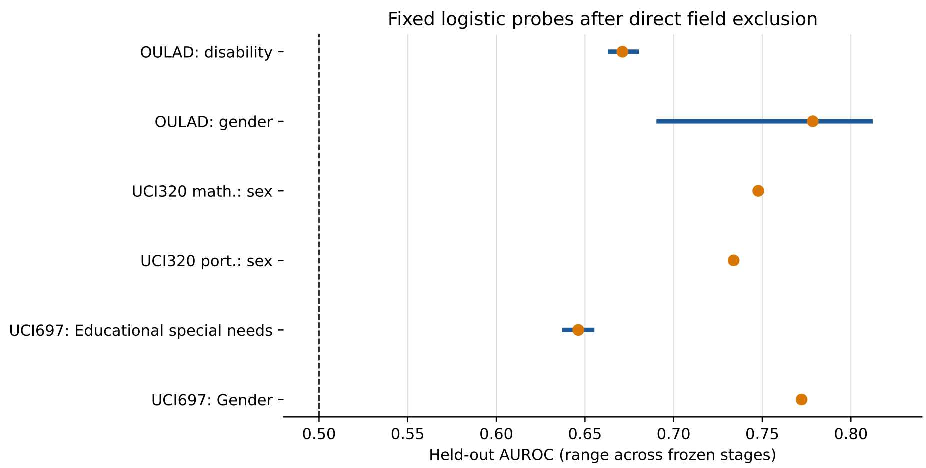

Figure 3. Held-out AUROC ranges for fixed logistic probes after direct field exclusion. Each segment spans the frozen stages and the point is their mean. The dashed line is chance discrimination. Predictability is not evidence that the outcome model actually used a proxy.

3.10. Harmonized blocks separated sex/gender from socioeconomic/family information

C14D assembled 30 exact same-run pairs (Table 8). No sex/gender workload-precision cell in OULAD, UCI 697, or UCI 320 had a family-wise interval excluding zero. This was not a general equality result: OULAD day-42 Brier score and UCI 697 enrollment AUROC showed small family-wise losses after gender exclusion, while UCI 320 Mathematics period-one AUROC showed a small improvement. Queue membership also remained imperfectly overlapping.

The socioeconomic/family block behaved differently. In OULAD, its AP and AUROC were lower and Brier score higher at every horizon after stage-wise family correction. Workload precision was below full after family-wise correction only at the 30% budget on days 28 and 56. In UCI 697, the block had a large enrollment penalty for both outcomes and a smaller but persistent semester-one penalty; all three global metrics favored full at both stages. UCI 320 again supplied a negative precision result: no workload cell excluded zero, and probability-metric directions were mixed. Figure 4 displays the distinct OULAD and UCI 697 budget profiles without pooling them.

Table 8. Primary harmonized precision families across all frozen stages and four budgets. Ranges are point estimates in percentage points; negative/positive columns count family-wise intervals excluding zero.

Context	Block	Cells	Point range	Negative	Positive
OULAD	sex/gender	24	[-0.23, +1.54]	0	0
OULAD	socioeconomic/family	24	[-4.05, +0.94]	2	0
UCI 320 Math.	sex/gender	12	[-5.00, +5.00]	0	0
UCI 320 Math.	socioeconomic/family	12	[-10.00, +5.04]	0	0
UCI 320 Port.	sex/gender	12	[-4.55, +2.94]	0	0
UCI 320 Port.	socioeconomic/family	12	[-4.55, +8.82]	0	0
UCI 697/Dropout vs all other	socioeconomic/family	8	[-21.78, -0.87]	7	0
UCI 697/Dropout vs graduate	sex/gender	8	[-1.09, +0.81]	0	0
UCI 697/Dropout vs graduate	socioeconomic/family	8	[-23.81, -1.35]	6	0

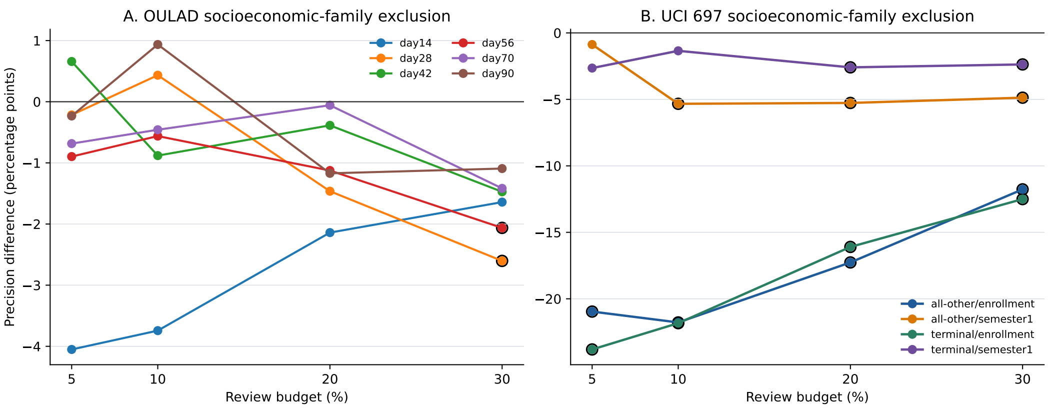

Figure 4. Socioeconomic/family exclusion across review budgets. Open black circles mark cells whose Bonferroni-percentile family-wise precision interval excludes zero. Lines connect estimates within a stage only. OULAD and UCI 697 use different populations, outcomes, and block definitions and are not pooled.

3.11. Multiplicity retained losses but did not create an equivalence claim

Across the combined C14B and C14D primary views, 296 stage–budget precision rows were evaluated. Thirty family-wise intervals were below zero and none was above zero. This count includes the planned policy and harmonized presentations of some exact same-run pairs; for example, UCI 697 no-family/financial and socioeconomic/family are numerically identical, so the duplicated rows are not independent confirmations. The corrected interpretation is therefore not that exclusion usually harms precision, but that concentrated losses survived in particular OULAD high-capacity queues and UCI 697 family/financial or sparse operational views, whereas sex/gender and all UCI 320 workload differences remained unresolved. An interval crossing zero is reported as inconclusive, not as proof of equivalence.

4. Discussion

4.1. Principal interpretation

The C14 stress test replaces a one-budget exclusion claim with a capacity-, stage-, block-, and metric-specific account. In OULAD, the original six-field policy passed its day-42 10% gate, yet minimized-model losses survived family-wise correction in several 30% queues. Sex/gender exclusion did not produce a family-wise workload loss, whereas removing the operational socioeconomic/family block reduced global predictive performance at every horizon and reduced workload precision in selected larger queues. Review capacity is therefore part of the estimand rather than a reporting convenience.

UCI 697 retained a large enrollment-time penalty for sparse operational and family/financial views across all four budgets. Stronger semester-one evidence attenuated those gaps but did not eliminate them at every capacity. UCI 320 constrains any generalization: G1 and G2 strongly improved the full controls, but neither harmonized block produced a family-wise workload difference in the small subject samples. It would be incorrect to attach the UCI 697 trajectory to UCI 320 or to call the latter an independent replication. The operational harmonization makes the contrasts auditable across datasets; it does not make the underlying constructs or tasks equivalent.

The difference between queue precision and global ranking metrics is substantive. Sex/gender exclusions could preserve capacity-constrained precision while AP, AUROC, or Brier showed small differences, and alert recipients changed even when aggregate precision was similar. Conversely, the socioeconomic/family block produced consistent global losses in OULAD without a workload loss at most queue boundaries. An institution must therefore specify the operating capacity and harm model, audit affected groups and individuals, and assess whether retained fields are necessary and proportionate.

The proxy audit adds a separate governance result. Directly deleted fields remained statistically predictable from the retained feature view, including among OULAD students not observed during probe fitting. This is compatible with redundant information and correlated social or behavioral processes, but it does not demonstrate that the outcome model encoded, relied on, or acted through a particular proxy. Direct exclusion alone neither removes statistical associations nor establishes fairness (Gardner et al., 2019; Pham et al., 2025). A stronger claim would require model-specific attribution or counterfactual tests, subgroup performance and allocation analyses, and a clearly specified normative criterion.

The strict-tier failure is also informative. Typed due-process features improved ranking relative to a sparse click-independent baseline but remained far below the standard comparator at the capacity boundary. A degraded mode should not be called secure merely because it excludes click inputs. Deployment would require documented data provenance and latency, platform threat analysis, access controls, monitoring, abstention or fallback rules, and prospective evaluation of the clean-utility cost.

4.2. Why provenance changed deterministic results

The audit demonstrates that a ZIP checksum is not administrative detail. A one-character missing-value representation altered preprocessing for a field used by full and partially excluded policies but not by the six-field-excluded policy. That pattern explains why the superseded results changed non-uniformly. Separately, changing an implicit row-order tie rule to an explicit SHA rule altered two day-28 boundary outcomes. Determinism applies only after raw bytes, parsing, missing-value semantics, feature construction, sorting, and software versions are fixed.

The correction procedure did not force C12 toward v0.2.1 or inherited C06/C08 counts. It used the official pinned bytes, retained the historical gates, reran every new policy with its same-run full control, and preserved all discrepancies and failed alternatives. This is necessary for auditability, but it does not erase earlier test-set exposure.

4.3. Relevance to educational practice

Module-specific queues approximate a local review budget better than one pooled institutional ranking. Precision, false alerts, recipient turnover, and degraded-mode utility can inform whether a system is ready for prospective study. They do not show that reviewing 10% is feasible for teachers, that alerts lead to appropriate action, or that intervention benefits students. Large-scale teacher-facing implementations and human-centred dashboard research show why workflow and interpretation must be studied with educators (Alfredo et al., 2024; Herodotou et al., 2019). Data-driven decision making also treats platform data as one input to contextual professional judgement (Evenstein Sigalov et al., 2026).

For governance and practice, the findings imply a staged deployment rule. Before removing or retaining a field block, an institution should (i) freeze the intended prediction time and the information genuinely available then; (ii) define the review capacity and same-run full comparator; (iii) evaluate workload, ranking, calibration, recipients, and subgroup outcomes; (iv) repeat the comparison when stronger process or grade evidence becomes available; and (v) specify abstention or fallback behavior when telemetry is unavailable. A teacher-facing dashboard should explain why a case entered the queue, permit professional override, record follow-up actions, and be evaluated prospectively for workload, decision quality, and student benefit. This is how the study connects to the quality of professional pedagogical activity: it tests the reliability and governance of a digital decision-support instrument, not the causal effectiveness of teachers themselves.

4.4. Limitations

First, all C12 analyses are post-development. The 2014J outcomes were exposed before the final protocol, the practical reference was selected after earlier results, and diagnostic official-data fits for days 14–56 were observed before the complete production plan was frozen. Bonferroni intervals address only the stated horizon family; they do not account for all policies, metrics, datasets, and prior decisions.

Second, the bootstraps are conditional on fitted pipelines and fixed selection and calibration choices. They do not capture model-selection, calibration, dataset, or development-process uncertainty. Some OULAD students contribute multiple registrations, while each UCI row was treated as one sampling unit. No institution-level transport sample was available.

Third, the outcome combines Fail and Withdrawn, which have different timing and possible intervention pathways. Risk sets change as withdrawals occur. Conservative assessment rules prevent use of scores with unknown release times but cannot reconstruct the actual availability time of every process record. Literal `?` handling reproduces the pinned recipe, but alternative semantically justified missing-data models warrant prospective specification.

Fourth, the common cohorts condition on later retention and cannot identify a causal timing effect. UCI 697 is a higher-education terminal-status task, whereas UCI 320 concerns subject grades in two secondary schools; neither matches OULAD's platform, outcome, or temporal split. The $G3 < 10$ threshold was frozen from the educational grading interpretation, not optimized, and Mathematics and Portuguese were not pooled. The strong G1/G2 result reflects predictive availability, not evidence that waiting for later grades is an optimal intervention policy.

Fifth, field exclusion does not eliminate proxies, and descriptive group audits cannot certify fairness. No analysis measures privacy attacks, teachers' workload in practice, intervention uptake, harm, or causal student benefit. Multiple metrics, policies, datasets, and prior development decisions remain beyond the within-policy horizon corrections.

Sixth, the C14 probe targets are observed dataset fields with dataset-specific coding. Probe discrimination is conditional on the frozen retained view and does not quantify sensitive-information leakage in an information-theoretic sense. The UCI 697 educational special-needs probe has only 40 positive cases, making its apparent lift especially unstable. The harmonized socioeconomic/family block combines area deprivation and prior education in OULAD, parental and current financial indicators in UCI 697, and parental education and occupation in UCI 320; its common label must not be mistaken for measurement invariance.

Seventh, the C14 family-wise intervals control only the prespecified cells within each named policy or block and metric. They do not form a single study-wide correction over every dataset, model family, proxy target, historical analysis, and narrative comparison. Exact C14B/C14D duplicate views are retained for traceability but are not independent evidence. Bootstrap intervals remain conditional on the fitted pipelines and do not incorporate hyperparameter-selection or transport uncertainty.

Finally, XuetangX was formally disqualified from demographic-exclusion evidence rather than left as an unresolved supporting dataset. The official KDD Cup 2015 files comprise enrollment identifiers, usernames and course identifiers; event logs; course dates and objects; and dropout labels, but no demographic fields. The internally preserved derivative that supplied demographics cannot be independently connected to that official schema. Canonical C05/v0.3.1 was preserved and not retrained, but it is excluded from the article's evidentiary claims. Inherited C06/C08 OULAD summaries also lack the complete models and predictions required for verification and remain `NOT_REVERIFIED`.

5. Conclusion

Data-minimization claims in educational early warning are not portable across fields, metrics, stages, or review capacities. Multi-budget analysis exposed OULAD losses that the 10% queue alone did not reveal. Sex/gender exclusion produced no family-wise workload penalty in the three datasets, but directly excluded fields remained predictable and small global-metric differences persisted. Socioeconomic/family exclusion reduced global performance throughout OULAD and UCI 697 and produced concentrated workload penalties, while UCI 320 remained inconclusive. Institutions should freeze the prediction time, review capacity, feature block, and same-run comparator; retain provenance and duplicate views transparently; audit ranking, calibration, recipients, and residual associations; and evaluate teacher workflow and student benefit prospectively. The present post-development evidence supports disciplined governance of digital decision support. It does not prove equivalence, fairness, privacy, or that alerts improve professional pedagogical practice or student outcomes.

Supplementary Materials: The following supporting information can be downloaded at https://www.mdpi.com/article/10.3390/educsci1010000/s1: Table S1: Computational evidence layers retained during the correction; Table S2: Primary restricted-minus-full precision families across all frozen stages and budgets; Table S3: Harmonized restricted-minus-full precision at a 10% review budget; Table S4: Primary fixed logistic proxy probes; Table S5: Official-data main-policy point estimates; Table S6: UCI 697 workload-policy contrasts; Table S7: UCI 320 workload-policy contrasts; Table S8: Leave-one-field-out minus full Precision@10% contrasts; Table S9: Common-cohort day-42-minus-day-14 difference-in-differences; Table S10: Candidate versus full alert-list identity; Table S11: True-alert transition from the superseded derivative-input run to C12; Table S12: Inherited NOT_REVERIFIED true-alert counts versus C12; Table S13: C12–C14 production and verification inventory. Supplementary Materials S1 is prepared as a separate MDPI supplementary-file-type document (see `supplement_mdpi.tex`) and is paginated and numbered independently (Table S1, Figure S1, ...) from the main text, per journal convention. Citations and references appearing in the Supplementary Materials also appear in the reference list of this main article.

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Informed Consent Statement: Not applicable. This study used only pre-existing, publicly available, de-identified secondary data; no human subjects were recruited or contacted by the authors.

Data Availability Statement: OULAD is available from the UCI Machine Learning Repository at https://archive.ics.uci.edu/dataset/349/open+university+learning+analytics+dataset and is described by Kuzilek et al. (2017). The official archive used here had SHA-256 `f2ed1902616c1fe8d2824d872c0b7d2d72be435bf0124d077044fe4be2c6d3e4`. UCI 697 is available at https://archive.ics.uci.edu/dataset/697/predict+students+dropout+and+academic+success; the outer ZIP used here had SHA-256 `e90e55fd65ec462ae283ebeb2cca409319e3460ed898d8754f62fb35cc83a65d`. UCI 320 is available at https://archive.ics.uci.edu/dataset/320/student+performance; the outer ZIP had SHA-256 `82ae9d66437b9808df42e8c89d2bb179c46e9cfbcf06f38abc1d20b3b747e177`. The current nested `student.zip`, not `.student.zip_old`, was analyzed. All three UCI pages identify the datasets as CC BY 4.0. The aggregate-only reproducibility release accompanying this submission is available at https://github.com/KuznetsovKarazin/SecureEWS; it contains the C14 analysis code, frozen protocol, machine-readable aggregate results and bootstrap draws, verification reports, and manuscript sources. Raw or processed student-level data, individual predictions, fitted model bundles, private clean-room archives, and XuetangX derivative files are not redistributed.

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